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Application of Microseismic and Optical Fiber Joint Monitoring Technique in Hydraulic Fracturing

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Hydraulic fracturing technology is to inject high pressure fluid into reservoir to create artificial fracture, increase reservoir connectivity and increase single well production.

In the early stages of horizontal well staged multi-cluster fracturing technology, traditional monitoring methods were mainly relied upon, such as downhole pressure and temperature sensors, as well as surface pressure and flow monitoring. These methods can provide basic construction parameters, but their monitoring accuracy in aspects such as fracture morphology and flow conductivity is limited, making it difficult to meet the development needs of complex reservoirs.

In recent years, microseismic monitoring technology has been widely used in hydraulic fracturing monitoring of oil and gas reservoirs^[1]. It is the only monitoring technology that can image the geometry of fracturing fracture. With continuous technological progress, microseismic monitoring technology has been introduced into horizontal well fracturing monitoring and has become one of the widely used hydraulic fracturing monitoring technologies in unconventional reservoir stimulation in recent years. Currently, this technology has developed from single-well monitoring to multi-well joint and large-array monitoring, greatly improving the positioning accuracy of microseismic events and the accuracy of fracture characterization.

Introduction

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The positioning accuracy of microseismic monitoring technology has achieved a transformative breakthrough—evolving from early single-well monitoring (positioning error $>50\text{m}$) to multi-well collaborative, large-aperture surface arrays (error $<10\text{m}$). This leap is accomplished by deploying geophone arrays in adjacent wells combined with extensive surface receiver systems, enhanced by waveform cross-correlation and 3D velocity model calibration. These advances significantly reduce ray-path uncertainties, ushering microseismic event localization into the era of sub-ten-meter accuracy (as field-verified in the Marcellus shale region).

In dynamic imaging, leveraging high-density event capture capability (>100 events/minute) and real-time inversion algorithms enables instantaneous microseismic event localization. This approach not only precisely characterizes the spatial configuration of primary fractures (length, height, azimuth) but also facilitates continuous fracture network simulations based on microseismic events. It identifies both primary and secondary fractures while enabling fracturing engineering risk warnings and real-time field fracturing scheme optimization through anomaly detection in microseismic event locations.

The continuous development of distributed optical fiber sensing (DFOS) technology has made fine-scale monitoring near the wellbore possible. DAS can accurately monitor the effects of downhole bridge plug, perforation, ball throwing, temporary plugging, and quantitatively calculate the amount of liquid inflow and sand inflow in a single cluster by monitoring the sound wave signal around the wellbore during hydraulic fracturing stimulation construction. DTS can be used to monitor downhole fluid production status in real time.

The two methods, microseismic and optical fiber, can combine with each other in the process of fracturing monitoring and have complementary advantages. It is possible to accurately grasp the fracturing effect around the wellbore through Optical fiber monitoring, and combined with the results of microseismic monitoring of fracture network size, it can better assist in adjusting and optimizing on-site fracturing plans

Method and Theory

Microseismic monitoring

The main method to do microseismic monitoring is to arrange geophones on the surface or downhole, passive data acquisition, locate and analyse microseismic events generated during hydraulic fracturing, to evaluate fracturing effects (Fig.1). The foundation is seismology, and the core is to calculate event parameters such as location, occurrence time and magnitude based on the seismic signals. According to the spatial distribution of microseismic events, the general steps of microseismic monitoring technology for hydraulic fracturing are as follows:

- Make microseismic acquisition plan according to the needs of monitoring tasks and objective conditions;
- Processing and interpreting real-time acquisition data, including geophone orientation, first arrival pickup, establishing speed model, etc., to determine the location, time and energy of microseismic events;
- According to the spatial distribution of microseismic events, the geometric parameters of fracturing cracks are determined, including the length, width and height of fracturing cracks and the volume of reservoir reconstruction

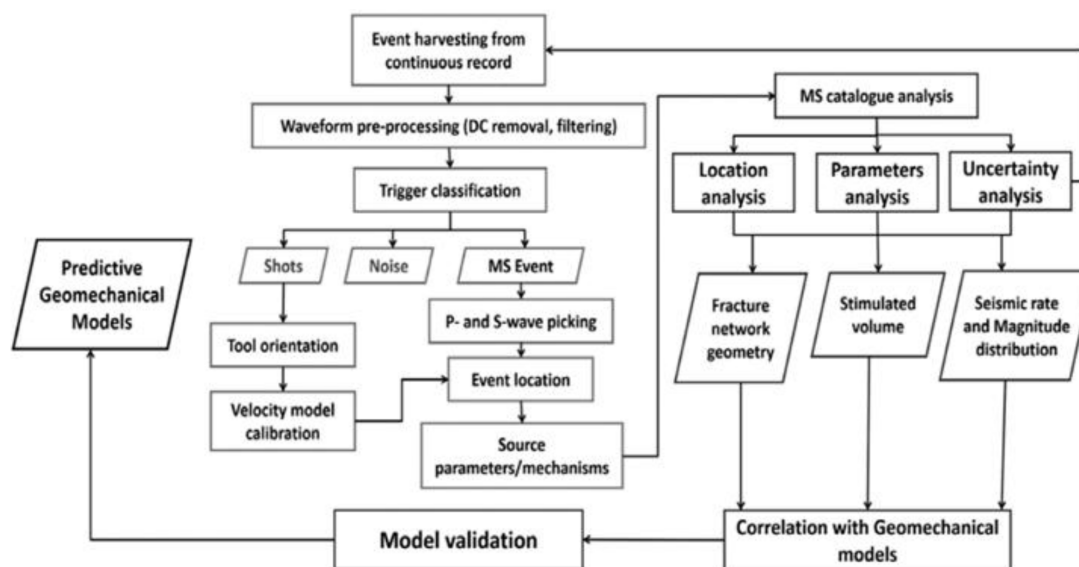


Figure 1—Microseismic event processing and interpretation workflow.

Distributed fiber optical sensing monitoring

As an emerging geophysical technology, the distributed optical fiber technology is most widely used in the oilfield field for three physical quantities: DAS (Distributed Acoustic Sensing), DTS (Distributed Temperature Sensing), and DSS (Distributed Strain Sensing). All three are based on different scattering of optical fiber, and there are differences in scattering frequency, energy, and other aspects. Therefore, the optical fiber technology can achieve applications in multiple scenarios.

DAS (Distributed Acoustic Sensing): Primarily monitors acoustic and vibration signals propagating along optical fibers. By utilizing the time-domain or phase information of Rayleigh scattering (Rayleigh Incident light), it enables the detection and localization of dynamic events such as microseismic activities, mechanical vibrations, and fluid disturbances.

DTS (Distributed Temperature Sensing): Relies on the Raman scattering effect in optical fibers. Raman scattered light consists of two frequency-shifted components: Stokes and Anti-Stokes, with their intensity ratio exhibiting an exponential relationship with temperature. By measuring the power ratio of these two scattered lights, DTS can real-time acquire temperature distribution information along the fiber. It provides long-range, continuous temperature profile monitoring with high precision and fast response speed.

DSS (Distributed Strain Sensing): Quantifies minute strains on optical fibers by detecting frequency shift changes in Brillouin scattering signals. This technology can real-timely reflect the strain distribution in structures, pipelines, or formations, making it critical for monitoring deformations in oil well bores, pipelines, and hydraulic fractures.

With the deepening application and widespread promotion of distributed optical fiber sensing technology in the oil and gas exploration and development field^[2-3], it will provide a new means for constructing smart oil and gas fields and achieve intelligent perception of oil and gas reservoir. Currently, optical fibers have the capability for integrated full-life-cycle monitoring covering drilling, cementing, fracturing, and production, assisting in optimizing on-site development plans and production regimes. It is expected to play an important technical supporting role in the future construction of smart oil and gas fields^[1].

During hydraulic fracturing, how to effectively achieve an accurate estimation of proppant and fluid injection into each fracturing cluster is of utmost importance. As an emerging monitoring tool, DAS technology can real-time monitor the acoustic signals of perforation clusters during the fracturing process, thereby inferring the proppant and fluid injection status of each fracturing cluster. Monitoring Basic Principle: When fracturing fluid enters the formation through perforation holes under high pressure, it

generates obvious pulsed acoustic responses on the outer wall of the casing due to factors such as fluid jetting, impact, turbulence, and local vibrations. Compared with conventional background noise, such signals differ in amplitude, attenuation rate, and spectral distribution, enabling the calculation of fluid injection volume into perforation clusters. Based on the results of experiments and computational fluid dynamics (CFD) simulations, assuming the impact of erosion effects and perforation phase on flow velocity is negligible, there is a significant correlation between acoustic energy and fluid flow velocity^[7], as shown in Equation (1), i.e., a linear relationship between acoustic energy and the logarithm of the cubic power of fluid flow velocity.

$$\log(q^3) = A \cdot L_{sp} + B$$

L_{sp} : Acoustic signal (sound pressure level), dB, q : Flow rate, in bbl./min, A , B : Calibration parameters, constants.

The energy of distributed optical fiber acoustic signals maintains the same correlation as acoustic signals and can sequentially replace L_{sp} in Equation (1). During hydraulic fracturing, the collected DAS acoustic data need to undergo digital filtering and fast Fourier transform (FFT) to convert the signals into the frequency band domain (FBE)^[6], and calculate the acoustic energy at specific depth positions within a certain time period. Further time-averaged FBE is used to determine the boundaries between different perforation clusters, thereby defining the initial depth windows. These windows are used to subsequently average the FBE signals of each perforation cluster, as shown in Equation (2).

$$E_{av}(x) = \frac{1}{N_t} \sum_{t=1}^{N_t} E(x, t)$$

E_{av} : Time-averaged value of FBE signals, in dB, N_t : Number of time steps during fracturing, $E(x, t)$: FBE frequency band energy at a certain time, in dB.

Using the above fluid injection estimation model, the FBE signal of each perforation cluster can be converted into fluid injection volume. To evaluate the fluid volume distribution throughout the fracturing stage, the energy of each perforation cluster needs to be accumulated over time to obtain the fluid injection volume of each cluster. Combined with fracturing operation parameters, this enables comprehensive analysis of proppant and fluid injection effects and guidance for fracturing operation optimization.

Concept of joint monitoring

In the process of hydraulic fracturing, DAS monitors the fine fracturing effect around the wellbore. Microseismic monitoring technology can describe the overall shape of the fracture network from the shape of event clouds. DTS temperature monitoring establishes the formation temperature waterfall diagram, roughly determines the transformation scope, compares and corrects the error generated by DAS monitoring.

Table 1—Table captions should begin with a short description of the table. Format tables using the Microsoft Word table commands and structures. Do not create tables using spaces or tab characters. Large tables presenting rich data should be presented as separate Excel or.csv files, not as part of the main text.

Table 1—Complementary analysis table of microseismic and fiber optic monitoring technologies.

Dimension	Microseismic Monitoring	Optical Fiber Monitoring	Combined Advantages
Spatial Coverage	Far-field fracture imaging	Fine responses near the wellbore	Achieve "far-field + near-field" full-scale coverage
Parameter Type	Dynamic event location	Continuous physical field monitoring (temperature/ acoustic/strain fields)	Integrate discrete event and continuous field evolution information
Diagnostic Depth	Answer "where the fractures extend"	Answer "how the fractures open/ how the fluid distributes"	

Case study

Two casing-outer optical fibers were deployed at the 3H platform in the Ganchaigou Formation in the Qinghai Oilfield (Fig.2). These fibers are used to evaluate the opening efficiency of stages/clusters, fluid injection volume, and temporary plugging effects, optimize horizontal well fracturing construction plans in real time, assess the fracturing effectiveness of deep shale oil in Qinghai, and explore the main controlling factors of post-fracturing production.

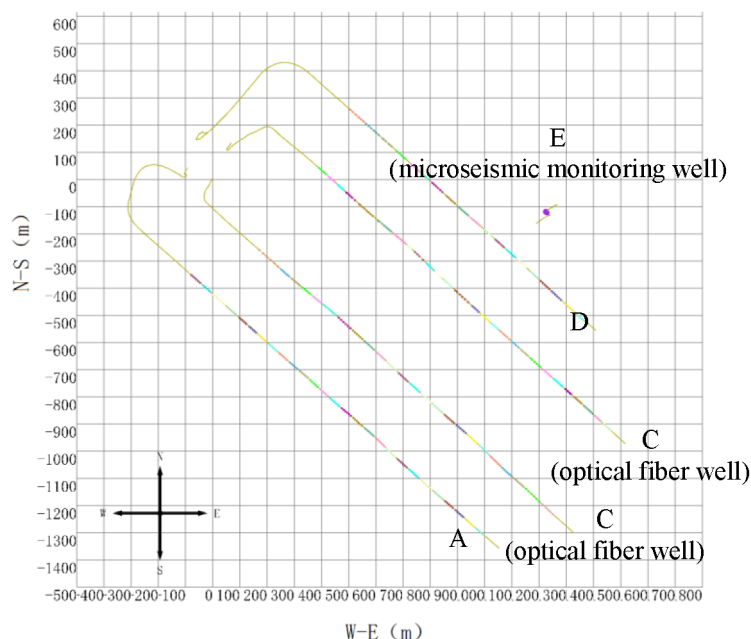


Figure 2—Microseismic and fiber optic monitoring geometry of 3H platform.

Among them, Well C has a total depth of 5,176.0 meters and a horizontal section length of 1,552.0 meters. The reservoir lithology is dominated by laminated and bedded dolomitic limestone, characterized by moderate-to-strong stress sensitivity, moderate-to-weak water sensitivity, weak alkali sensitivity, and weak salt sensitivity. In terms of physical properties, the horizontal section of Well C has an average Young's modulus of 41.4 GPa, an average Poisson's ratio of 0.296, an average maximum horizontal principal stress of 95.9 MPa, an average minimum horizontal stress of 80.5 MPa, an average horizontal stress difference of 15.4 MPa, and an average fracturability index of 0.52, indicating good overall formation fracturability.

Based on the characteristics of shale oil reservoirs, a dense cutting volume fracturing strategy was adopted to form complex fracture networks. A total of 27 stages and 148 clusters were designed for fracturing, with a total fluid injection volume of 47,500 m³, total proppant volume of 4,696 m³, fluid injection intensity of 32 m³/m, proppant intensity of 3.16 m³/m, and single-stage proppant volume ranging from 130 to 250 m³.

Meanwhile, we selected offset well E adjacent to the platform as the monitoring well for microseismic monitoring, and deployed 12-level downhole geophones within it.

Optimized stage perforation pattern

In Well C, during fracturing, high-precision vibration, and temperature change monitoring around the wellbore by optical fiber are used to reflect the opening of each cluster (Fig.3) and the proppant and fluid injection effects of each stage/cluster were monitored via optical fibers. The red line indicates the depth of the bridge plug in the wellbore. The interval above the bridge plug represents the target fracturing stage in this operation, while the interval below the bridge plug corresponds to the previous fracturing stage.

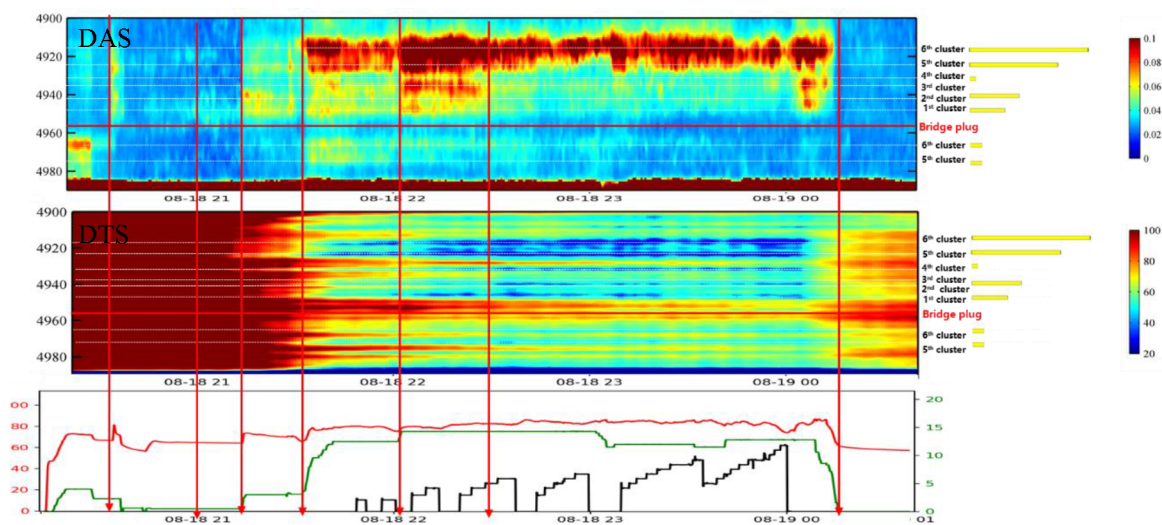


Figure 3—Use DAS/DTs waterfall plot to jointly analyse the effect of sand injection in each cluster.

During the early fracturing stage, uniform perforation fracturing was applied. As shown in Fig.4(a), the intra-stage opening efficiency was low, with fracturing fluid primarily concentrated at the heel end of each stage (i.e., the closer end to the wellbore), while the opening and fluid injection effects at the toe end (i.e., the farther end from the wellbore) were relatively poor. The uneven fluid distribution within the stage led to unbalanced propagation of artificial fractures, affecting the overall fracturing performance.

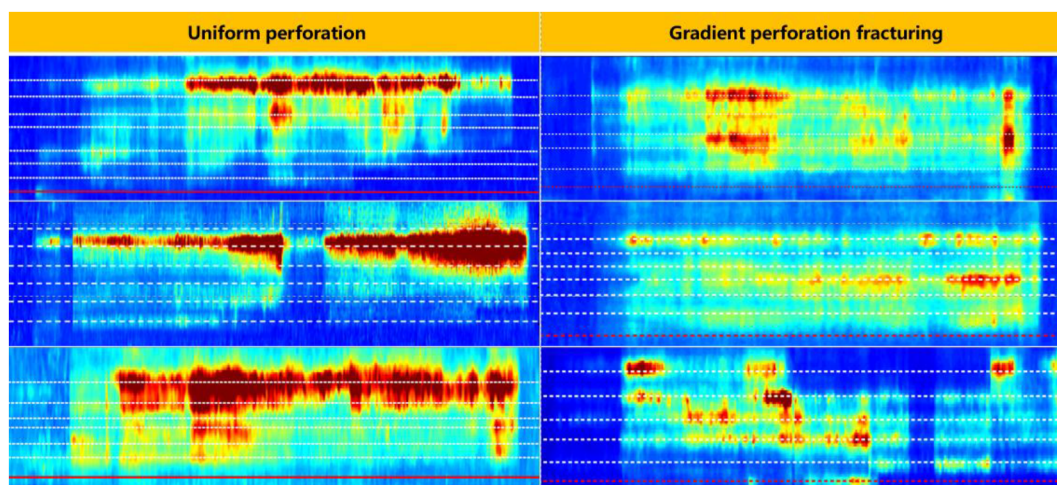


Figure 4—Comparison of Optical Fiber DAS Signals Between Uniform Perforation Fracturing and Slanted Perforation Fracturing (L) Uniform Perforation (R) Slanted Perforation

To improve reservoir stimulation effectiveness, the perforation method was adjusted in real time on site, adopting gradient perforation fracturing as shown in Fig. 4(R). After the adjustment, the opening and fluid injection effects of clusters at the toe end were significantly improved. Compared with uniform perforation, the cluster opening efficiency increased after gradient perforation, allowing fracturing fluid to enter the formation more uniformly and significantly enhancing the stimulation uniformity of each stage.

According to the actual monitoring results of Well C, uniform perforation failed to achieve the effect of uniform initiation in each cluster. The site promptly adjusted the perforation method by reducing the number of perforating charges at the heel end of the stage and increasing them at the toe end. This optimization of the perforation strategy improved the fracturing stimulation effect and provided strong data support for the optimization of perforation schemes in this area in the future.

Optimized bridge plug construction method

Staged multi-cluster fracturing in horizontal wells is an effective means to improve the production of oil and gas wells. Bridge plugs are key tools in this process, providing a relatively independent fracturing environment for each stage of fracturing operations to ensure that fracturing fluids and proppants can better enter the formation, thereby forming complex artificial fractures. Therefore, the sealing performance of bridge plugs is directly related to the quality of fracturing results.

During the optical fiber fracturing monitoring of Well C, on-site monitoring of the sealing effect of bridge plugs was carried out, and the fluid intake and leakage volume of the current stage were calculated as shown in Fig. 5. The red dashed line in the Fig. indicates the position of the bridge plug. It can be seen from Fig. 5(L) that the leakage rates of these two stages during fracturing were 20% and 17% respectively. Field statistics show that the downhole times of the bridge plugs in these two stages were 11.5 hours and 11 hours respectively. Field analysis indicates that bridge plug leakage is related to the long downhole time of bridge plugs. As shown in Fig. 5(R), fracturing was carried out within 2.5 hours and 2.2 hours after the bridge plugs were put into the well, and the leakage rates monitored by optical fibers were 0% and 8% respectively. The comparison shows that the shorter the downhole time of the bridge plug, the better the sealing performance.

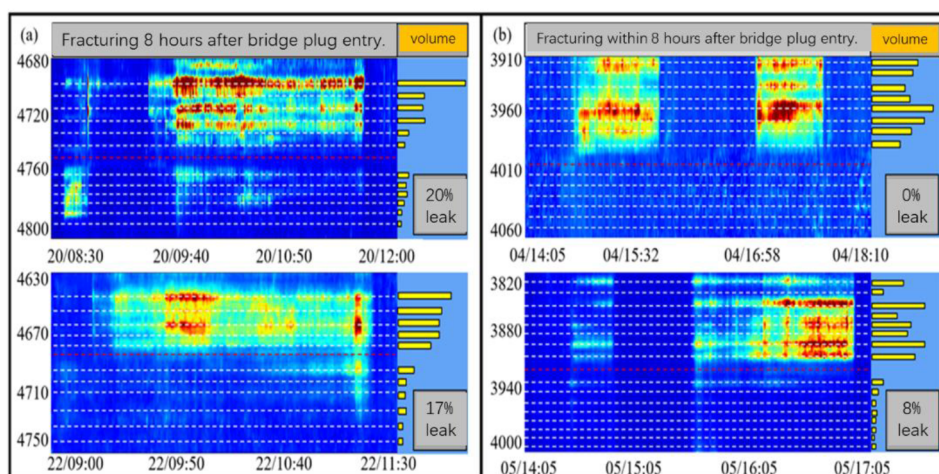


Figure 5—Comparison of Leakage phenomena of fracturing within and after 8 hours of plug setting in well C

Statistics on the sealing of bridge plugs in each stage of Well C show that when fracturing is carried out more than 8 hours after the bridge plug is put into the well, the sealing performance is poor, with an average leakage rate of 14.25%; when fracturing is carried out within 8 hours after the bridge plug is put into the well, the sealing performance of the bridge plug is good, with an average leakage rate of 6.82%, which ensures the opening of the fracturing clusters in the current stage.

Microseismic monitoring

The 3H platform was designed for 97 fracturing stages, with a total of 88 stages monitored and 3,372 monitoring event points collected. Limited by the monitoring range, Wells A and B collected fewer event points. Microseismic monitoring results from Wells C and D show inter-well fracture network communication and adequate inter-stage stimulation, indicating that the overall stimulation performance of the platform is positive.

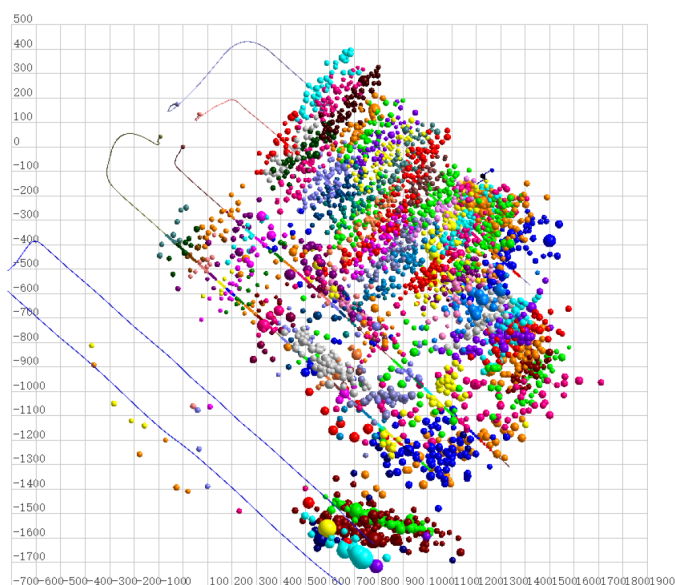


Figure 6—Map view of microseismic monitoring of platform 3H

Judging the interconnection with natural fractures based on microseismic monitoring

During real-time monitoring, the occurrence location and magnitude of microseismic events, B-value, and other parameters are calculated by observing changes in oil pressure. It is found that the natural fracture zone is communicated during the fracturing process, with numerous high-magnitude events occurring. Their orientations are not affected by the direction of principal stress but extend along the direction of the natural fracture zone. Additionally, the pressure decreases significantly during fracturing. The shape of the fracture network matches well with the natural fracture zone indicated by the curvature attribute.

During the fracturing of stages 10-13 in Well C, abnormal signals appeared near the wellbore of stages 10-19 in Well B (Fig. 7). It was determined that natural fractures might exist in stages 10-19 of Well B, posing a casing damage risk. Site construction risk warnings were issued, noting that subsequent perforation and fracturing operations in stages 10-19 of Well B should pay attention to construction risks. It is recommended to reduce the construction scale in subsequent operations. During the perforation operation of stages 10-19 in Well B, blockages were encountered at multiple depths, including 4334m, 4333m, 4332m, 4185m, 4190m, and 3696m. Comprehensive analysis indicates that natural fractures around Well B were activated during the fracturing of Well C, causing casing deformation, which is consistent with the location of micro-seismic monitoring and early warning.

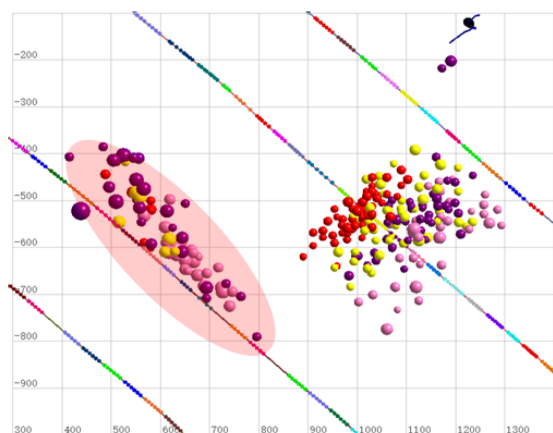


Figure 7—Map view of induced microseismic event in stage 10-19 of well B

Description of fracture network geometry

A total of 18 stages were monitored by microseismic in Well C. Based on the distribution of microseismic event points, the geometric parameters of the fracture network were described by elliptical fitting. Among them, Stages 3-6 were affected by natural fractures, with the north wing extending better than the south wing. The extension of the north and south wings in the remaining stages was relatively symmetrical, with an average of 196m for the north wing and 161m for the south wing. A total of 25 stages were monitored in Well 15-4, with relatively symmetrical extension of the north and south wings in each stage, with an average of 167m for the north wing and 178m for the south wing (Fig.8).

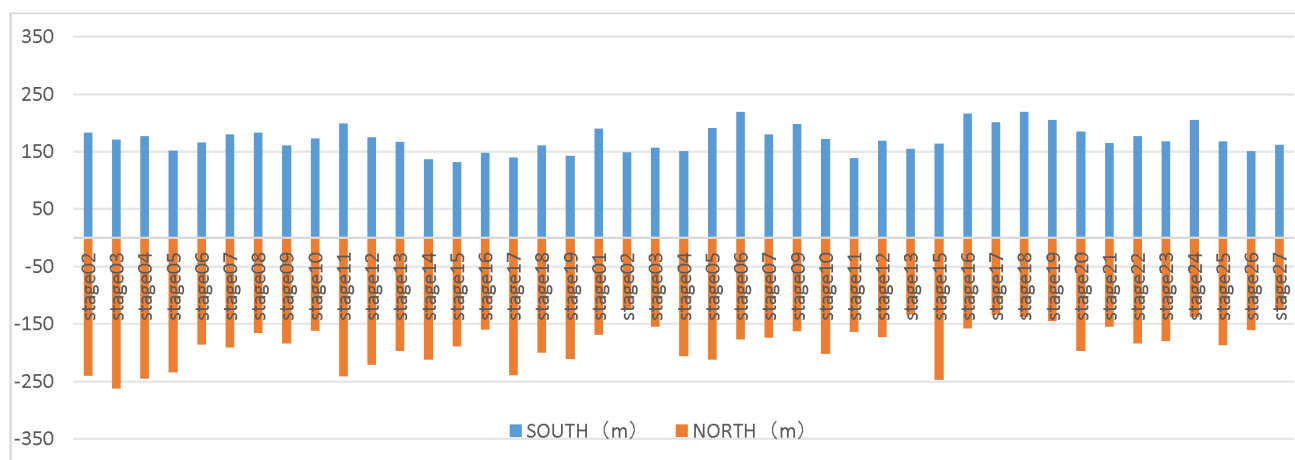


Figure 8—Statistically analyze the lengths of the southern and northern wings of each fracture stage based on microseismic.

Integrated analysis of fiber optic & microseismic monitoring

The fiber optic monitoring results of Stage 24 show that Clusters 1 and 6 had strong fluid intake before plugging. After plugging, the fluid intake of Clusters 1 and 6 decreased, while that of Clusters 2 and 5 increased. This is consistent with the microseismic monitoring results of this stage, which indicate that before plugging, fractures were generated in Clusters 1, 3, and 6 of this stage. After plugging, Clusters 2 and 5 were activated and extended, and new fractures were initiated in Cluster 6 (Fig 9).

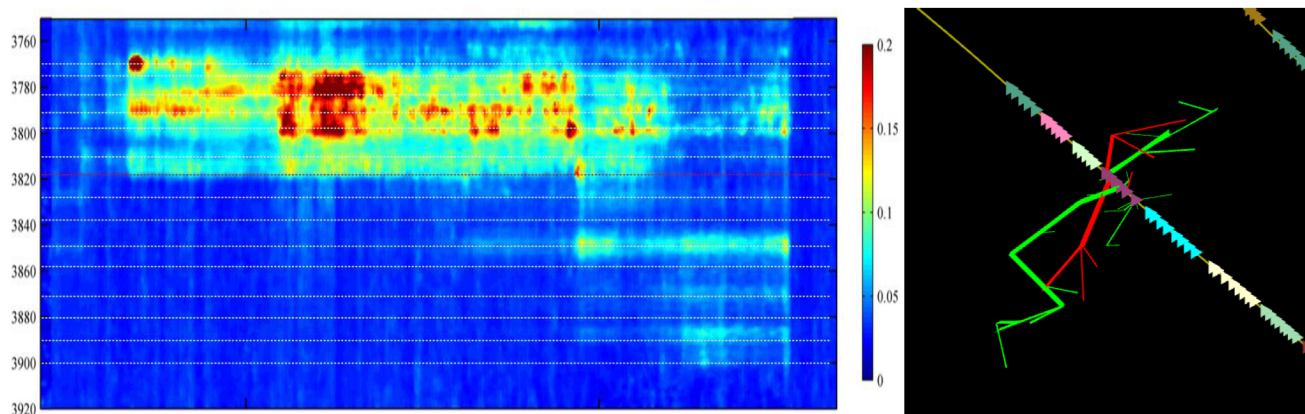


Figure 9—Fiber optic DAS waterfall plot (L) and microseismic continuous fracture network simulation results (R) for Stage 24.

When evaluating the effectiveness of temporary plugging, DAS/DTS are sensitive to vibrations and temperatures near the wellbore. They can visually judge the change in the inflow rate before and after plugging, and the application effect of the plugging ball reflects more strongly. Real-time monitoring of microseismic fractures can continuously observe the state of crack propagation, and analyse the distribution

of event points, SRV differences, and complexity changes before and after plugging. The two methods complement each other's strengths and weaknesses and can more accurately evaluate the effect of temporary plugging. Fiber optics are used to monitor ultra-low-frequency DAS strain in adjacent wells, and combined with the extension of microseismic events, to identify frac-hits and reduce engineering risks.

Conclusions

Each monitoring method has specific advantages and disadvantages. Microseismic monitoring excels in 3D fracture network characterization, providing critical insights into fracture geometry (length, height, complexity) and dynamic propagation. However, it lacks the ability to quantify fluid flow parameters at the cluster scale and may face challenges in signal clarity within complex reservoirs. In contrast, fiber optic monitoring offers unique advantages: DAS enables high-resolution identification of perforation cluster activation and precise flow quantification, resolving the ambiguity of multi-cluster flow distribution, while DTS provides continuous temperature field tracking, revealing fluid migration paths and reservoir thermal connectivity over time.

These monitoring methods can complement each other in the fracturing and production process of oil and gas wells. Through the establishment of downhole temperature and pressure system, the production status of oil and gas wells is monitored in real time, and the analysis of fracturing and production status is carried out by comprehensive use of temperature, micro-seismic and acoustic data to provide guidance for the formulation of next production measures.

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